Day 1: Multi-Digit Place Value & Number Modelling

Curriculum Reference: B4.1.1.1 • **Indicator:** B4.1.1.1.1 • **Standard:** International British Mathematics

1. Foundational Preamble: Digits vs Numbers

Before working with multi-digit quantities, we must establish the precise mathematical boundary between a **digit** and a **number**:

Term	Mathematical Definition	Everyday Analogy
Digit	A single mathematical symbol used to build values. In the decimal system, there are exactly ten digits: **0, 1, 2, 3, 4, 5, 6, 7, 8, 9**.	Like individual letters in the alphabet (e.g. *C, A, T*).
Number	A complete quantity or count formed by combining one or more digits.	Like a full word formed by letters (e.g. *CAT*).

2. The Base-10 Mechanism: The Power of Ten

Our numbering structure is a **Base-10 positional system**. Whenever ten items accumulate in any given column, they bundle together to create **one** item in the immediate next column to the left:

The Core Law of Place Value:
• Moving **one column left** multiplies the value by **10**.
• Moving **one column right** divides the value by **10**.
• 10 Units = 1 Ten | 10 Tens = 1 Hundred | 10 Hundreds = 1 Thousand | 10 Thousands = 1 Ten Thousand | 10 Ten Thousands = 1 Hundred Thousand.

3. The Standard 6-Column Place Value Chart

Large quantities are organised into 3-digit groups known as **Periods**. In British conventions, periods are separated by a comma (e.g. **73,425**):

Thousands Period			Units Period		
Hundred Thousands (HTh)	**Ten Thousands (TTh)**	**Thousands (Th)**	**Hundreds (H)**	**Tens (T)**	**Units (U / Ones)**
100,000	**10,000**	**1,000**	**100**	**10**	**1**
-	**7**	**3**	**4**	**2**	**5**

4. Distinguishing Between 'Place' and 'Value'

Inspect the numeral **73,425** closely:

- **Place:** The column/position where the digit sits (e.g. Ten Thousands).
- **Value:** The numerical worth of the digit due to its place (e.g. 7 in Ten Thousands place = 70,000).

Digit	Place Name	Calculation	Total Value

7	Ten Thousands (TTh)	7 x 10,000	70,000
3	Thousands (Th)	3 x 1,000	3,000
4	Hundreds (H)	4 x 100	400
2	Tens (T)	2 x 10	20
5	Units (U)	5 x 1	5

5. Expanded Form & The Role of Zero as a Place Holder

Expanded Form (Partitioning): Expressing a number as the addition of each digit's value:

73,425 = 70,000 + 3,000 + 400 + 20 + 5

When zero appears within a multi-digit number, such as **60,408**, it holds the column open so other digits retain their correct value. Without zero, 60,408 would collapse into 648. Thus, 60,408 = 60,000 + 400 + 8.

6. Guided Practice & Checkpoint Questions

Question 1: In the number **84,529**, what is the place of digit 8, and what is the value of digit 4?
Answer: Digit 8 sits in the Ten Thousands place. Digit 4 has a value of 4,000.

Question 2: Write **90,604** in expanded notation.
Answer: 90,000 + 600 + 4.

Question 3: What standard numeral is formed by **50,000 + 8,000 + 300 + 70 + 2**?
Answer: 58,372.

Question 4: How many groups of 1,000 are in **100,000**?
Answer: Exactly 100 groups of 1,000.

Question 5: What happens to a digit's value when it moves two places to the left?
Answer: It becomes 100 times larger (10 x 10).